

FINAL REPORT

Test Facility Study No. 9601567

SM-102
Bacterial Reverse Mutation Test in Salmonella Typhimurium and
Escherichia Coli

SPONSOR:

Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139
USA

TEST FACILITY:

Charles River Laboratories Montreal ULC
(b) (6)



12 January 2017

Page 1 of 75

TABLE OF CONTENTS

LIST OF TABLES	3
LIST OF APPENDICES	4
COMPLIANCE STATEMENT	5
QUALITY ASSURANCE STATEMENT	6
1. RESPONSIBLE PERSONNEL	7
1.1. Test Facility	7
1.2. Individual Scientists (IS) at Test Facility	7
2. SUMMARY	8
3. INTRODUCTION	9
4. MATERIALS AND METHODS	9
4.1. Test and Reference Items	9
4.1.1. Test Item	9
4.1.2. Reference Items	9
4.2. Test Item Characterization	10
4.3. Analysis of Test Item	10
4.4. Test Item Inventory and Disposition	10
4.5. Dose Formulation and Analysis	10
4.5.1. Preparation of Reference Items	10
4.5.2. Preparation of Test Item	11
4.5.3. Sample Collection and Analysis	11
4.6. Test System	12
4.6.1. Justification for Test System and Dose Level Selection	12
4.7. S9 Mix	12
4.8. Sterility Check and Spontaneous Mutation Rates	12
4.9. Plate Incorporation Method	13
4.10. Experimental Design	13
5. COMPUTERIZED SYSTEMS	13
6. EVALUATION AND INTERPRETATION OF RESULTS	14
6.1. Assay Acceptance Criteria:	14
6.2. Interpretation of Results	15
7. RETENTION OF RECORDS	15
8. RESULTS	16
8.1. Dose Formulation Analyses	16
8.2. Bacterial Mutation Test	16
9. CONCLUSION	17

LIST OF TABLES

Table 1 SM-102 - Plate Incorporation Assay in the Absence of S9 Mix.....18

Table 2 SM-102 - Plate Incorporation Assay in the Presence of S9 Mix20

Table 3 Positive Controls for the Plate Incorporation Assay22

LIST OF APPENDICES

Appendix 1 Study Plan, Amendment and Deviations23

Appendix 2 Test Item Characterization53

Appendix 3 Historical Negative and Positive Control Results.....56

Appendix 4 Dose Formulation Analysis Report.....58

COMPLIANCE STATEMENT

The study was performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA), Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions from the above regulations are listed below.

- Characterization of the Test Item was performed by the Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses were not conducted in compliance with the GLP or GMP regulations.
- Stability testing of the supplied Test Item was not determined in this study. It will be performed by the Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

This study was conducted in accordance with the procedures described herein. All deviations authorized/acknowledged by the Study Director are documented in the Study Records. The report represents an accurate and complete record of the results obtained.

There were no deviations from the above regulations that affected the overall integrity of the study or the interpretation of the study results and conclusions.

(b) (6)

Date: 12 Jan 2017

QUALITY ASSURANCE STATEMENT

Study Number: 9601567

This Study has been audited by Quality Assurance in accordance with the applicable Good Laboratory Practice regulations. Reports were submitted in accordance with SOPs as follows:

QA INSPECTION DATES

Date(s) of Audit	Phase(s) Audited	Dates Findings Submitted to:	
		Study Director	Study Director Management
12-Sep-2016	Final Study Plan	22-Sep-2016	22-Sep-2016
12-Sep-2016	Dose Preparation	22-Sep-2016	22-Sep-2016
22-Sep-2016	Study Plan Amendment 1	22-Sep-2016	22-Sep-2016
01-Nov-2016 - 02-Nov-2016	Data Review - Analytical Chemistry	04-Nov-2016	04-Nov-2016
01-Nov-2016 - 02-Nov-2016	Final Phase Report - Dose Formulation Analysis	04-Nov-2016	04-Nov-2016
08-Nov-2016 - 09-Nov-2016	Data Review - In Vitro Sciences	09-Nov-2016	09-Nov-2016
08-Nov-2016 - 09-Nov-2016	Final Report	09-Nov-2016	09-Nov-2016

In addition to the above-mentioned audits, process-based and/or routine facility inspections were also conducted during the course of this study. Inspection findings, if any, specific to this study were reported by Quality Assurance to the Study Director and Management and listed as a Phase Audit on this Quality Assurance Statement.

The Final Report has been reviewed to assure that it accurately describes the materials and methods, and that the reported results accurately reflect the raw data.

(b) (6)

12-Jan-2017
Date

1. RESPONSIBLE PERSONNEL

1.1. Test Facility

Study Director

(b) (6)

Test Facility Management

(b) (6)

1.2. Individual Scientists (IS) at Test Facility

Dose Formulation Analysis

(b) (6)

2. SUMMARY

The objective of this study was to determine the potential genotoxicity of SM-102, using the bacterial reverse mutation test.

The experimental design was as follows:

Text Table 1
Plate Incorporation Assay

Dose No.	Formulation Conc. (µg/mL)	Dose Volume (µL/plate)	Final Conc. (µg/plate)	Number of Replicates		Number of Strains
				0S9	+S9	
Negative Control	-	100	-	3	3	5
1/ SM-102	19.0 ^b	83.2	1.58	3	3	5
2/ SM-102	58.2 ^b	85.9	5.0	3	3	5
3/ SM-102	184 ^b	85.9	15.8	3	3	5
4/ SM-102	500 ^a	100	50	3	3	5
5/ SM-102	1581 ^a	100	158	3	3	5
6/ SM-102	5000 ^a	100	500	3	3	5
7/ SM-102	15811 ^a	100	1581	3	3	5
8/ SM-102	50000 ^a	100	5000 ^c	3	3	5
Positive controls	d	100	d	3	3	5

a Theoretical concentration; actual concentration may differ slightly due to the limitations of the instruments used.

b Measured concentration

c Test Item was tested at levels up to 5000 µg/plate, which is the standard limit dose recommended by regulatory guidelines.

d Dose depends on the test organism, the positive controls and methodology used (see [Section 4.5.1.2](#))

Salmonella typhimurium strains (TA1535, TA1537, TA98, TA100) and *Escherichia coli* strain WP2 *uvrA* were treated with the Test Item at a range of concentrations up to 5000 µg/plate (the standard limit dose for this assay), in the presence and absence of a supplemented rat liver fraction (S9 mix), using the plate incorporation version of the bacterial reverse mutation test.

Bacteria were incubated with standard positive controls, and the response of the various bacterial strains to these agents confirmed the sensitivity of the test system and the activity of the S9 mix.

Incomplete, or absent, background lawns of non-revertant bacteria, or substantial reductions in revertant colony counts, were not obtained following exposure to SM-102, indicating that the Test Item was non-toxic to the bacteria at the levels tested. Precipitation was observed at concentrations ≥ 1581 µg/plate in the absence of S9 mix and at concentrations ≥ 500 µg/plate in the presence of S9 mix.

No substantial increases in revertant colony numbers were obtained with any of the tester strains, following exposure to SM-102 at any dose level, in either the presence or absence of S9 mix.

It is concluded that SM-102 did not show any evidence of genotoxic activity in this in vitro mutagenicity assay when tested in accordance with regulatory guidelines.

3. INTRODUCTION

The objective of this study was to determine the potential genotoxicity of SM-102, using the bacterial reverse mutation test.

The design of this study is based on OECD Guideline 471 and ICH Guideline S2(R1).

The Study Director signed the Study Plan on 08 Sep 2016, and dosing was initiated on 13 Sep 2016. The experimental start date was 12 Sep 2016, and the experimental completion date was 19 Sep 2016. The study was completed on the date of the Study Director approval of this report (refer to the appropriate signature page). The Study Plan, last Study Plan amendment, and deviations are presented in [Appendix 1](#).

4. MATERIALS AND METHODS

4.1. Test and Reference Items

4.1.1. Test Item

Identification:	SM-102
Batch (Lot) No.:	RL-100-211-1
Retest Date:	27 Oct 2017
Purity:	95.72% (All concentrations and dose levels throughout this report were corrected for purity using a purity of 95.3%.)
Storage Conditions:	Kept in a freezer set to maintain -20°C

4.1.2. Reference Items

4.1.2.1. Negative Control

Identification:	Ethanol
Supplier:	Commercial Alcohols
Batch /Lot No.:	020612
Expiration Date:	June 2017
Storage Conditions:	Kept at ambient room temperature

4.1.2.2. Positive Controls

In the absence of S9 mix:

Identity:	Sodium azide (NaN ₃)
CAS number:	26628-22-8
Identity:	9-Aminoacridine hemihydrate (9AC)
CAS number:	65944-23-2

Identity: 2-Nitrofluorene (2NF)

CAS number: 607-57-8

Identity: 4-Nitroquinoline N-oxide (NQO)

CAS number: 56-57-5

In the presence of S9 mix:

Identity: 2-Aminoanthracene (2AA)

CAS number: 613-13-8

Identity: Benzo[a]pyrene (BaP)

CAS number: 50-32-8

Full details of positive controls including supplier, lot number, storage, expiry date and formulation are retained as part of the Test Facility records. Copies of certificates of analysis are retained as raw data.

4.2. Test Item Characterization

The Sponsor provided to the Test Facility documentation of the identity, strength, purity, composition, and stability for the Test Item. A Certificate of Analysis was provided to the Test Facility and is presented in [Appendix 2](#).

4.3. Analysis of Test Item

A Certificate of Analysis was provided by the Sponsor and is presented in [Appendix 2](#).

4.4. Test Item Inventory and Disposition

Records of the receipt, distribution, and storage of Test Item were maintained with the study raw data. All unused Test Item was returned to the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

4.5. Dose Formulation and Analysis

4.5.1. Preparation of Reference Items

4.5.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, was dispensed into a vial for administration to control plates. The aliquot was stored in a refrigerator set to maintain 4°C until use. Any residual volumes were discarded before issuance of the Final Report.

4.5.1.2. Preparation of Positive Controls

The positive controls formulations were prepared up to 6 months prior to use. Adequate amounts were dispensed into vials, and stored in a freezer set to maintain -20°C, protected from light, until use. The aliquots were removed from the freezer and allowed to warm to ambient room temperature before dosing. Any residual volumes were discarded after completion of dosing.

Depending on the strains tested, the following positive controls were used:

Positive Controls for the Assay

Strain	S9	Positive Controls	Concentration µg/plate	Vehicle
TA1535, TA100	0	Sodium azide (NaN ₃)	0.5	Sterile water
TA1537		9-Aminoacridine hemihydrate (9AC)	50	DMSO
TA98		2-Nitrofluorene (2NF)	1	DMSO
WP2 <i>uvrA</i>		4-Nitroquinoline N-oxide (NQO)	0.5	DMSO
TA1535	+	2-Aminoanthracene (2AA)	5	DMSO
WP2 <i>uvrA</i>			20	
TA1537, TA98, TA100		Benzo[a]pyrene (BaP)	5	DMSO

4.5.2. Preparation of Test Item

The Test Item was prepared as a stock solution (50 mg/mL) in the chosen vehicle (ethanol) and all lower level formulations were made by serial dilution. The formulations were prepared 1 day prior to use and were stored in a refrigerator set to maintain 4°C until use. The formulations were removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes were discarded before issuance of the Final Report.

4.5.3. Sample Collection and Analysis

The positive control formulations were not subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Dose formulations 1 to 8 were collected for concentration analysis only. Homogeneity, density and stability were not determined on these samples.

Samples to be analyzed were transferred at ambient room temperature, to the Analytical Chemistry department at the Test Facility on the date prepared. Any residual/retained analytical samples were discarded before issuance of the Final Report.

4.5.3.1. Analytical Method

Analyses were performed by HPLC, using a validated analytical procedure (Test Facility Study Number 1801841).

4.5.3.2. Concentration Analysis

Duplicate 1 mL samples for dose numbers 1 and 2 and duplicate 0.5 mL samples for dose numbers 3 to 8 were taken and sent to the analytical laboratory for analysis. Additional duplicate 1 mL samples for dose numbers 1 and 2 and duplicate 0.5 mL samples for dose numbers 3 to 8 were taken and retained at the Test Facility as backup samples. Concentration results were considered acceptable if mean sample concentration results were within $\pm 10\%$ of theoretical for the stock solution and $\pm 15\%$ of theoretical for lower level solutions. After acceptance of the analytical results, backup samples were discarded.

4.5.3.3. Stability Analysis

Stability analyses performed previously as part of a separate GLP-compliant study (Test Facility Study Number 1801841) demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study.

4.6. Test System

The bacteria were originally supplied by Moltox, NC, USA. Each batch of frozen bacteria was tested for appropriate phenotype characteristics and spontaneous reversion rates; response to diagnostic mutagens is also routinely assessed. The following bacterial strains were employed:

S. typhimurium TA1535 *hisG46 rfa ΔuvrB*

S. typhimurium TA1537 *hisC3076 rfa ΔuvrB*

S. typhimurium TA98 *hisD3052 rfa ΔuvrB* pKM101

S. typhimurium TA100 *hisG46 rfa ΔuvrB* pKM101

E. coli WP2 *trp uvrA*

Fresh bacterial cultures were prepared so that they were in the late-log phase of growth at the time of use. The density of the cultures was confirmed to be $\geq 1000 \times 10^6$ bacteria/mL using a bacterial counting chamber before the cultures were used in the test.

4.6.1. Justification for Test System and Dose Level Selection

The bacterial reverse mutation test detects point mutations, which are the cause of many human genetic diseases and can play a role in tumor initiation and development.

Typically, the Test Item is dosed at a range of concentrations but is only assessed at the five highest levels below the toxic level or, if non-toxic, at five levels up to the standard limit of 5000 µg/plate.

4.7. S9 Mix

The S9 mix, used as a model of intact mammalian metabolism, was prepared on the day of use and contained 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM MgCl₂, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate and 4 mM NADP. The S9 mix was stored in a refrigerator set to maintain 4°C then held on ice during utilization. A copy of the manufacturer's quality control certificate for the S9 fraction is retained as raw data.

4.8. Sterility Check and Spontaneous Mutation Rates

The sterility of the Test Item formulation (high dose) was confirmed on the day of the test using appropriate preparations without bacteria.

The spontaneous mutation rates of the bacterial strains were assessed using concurrent control samples in which the bacteria were exposed to the Negative Control.

4.9. Plate Incorporation Method

A 0.5 mL aliquot of S9 mix (+S9) or phosphate buffer 0.2 M pH 7.4 (0S9) was combined with 0.1 mL bacterial culture in a sterile container. An aliquot of the test item (see [Text Table 2](#) for test item dose volumes) or a 0.1 mL aliquot of reference item was added, then 2 mL of molten top agar supplemented with 0.05 mM biotin and minimal (0.05 mM) histidine and minimal (0.05 mM) tryptophan was added immediately afterward. The solution was mixed and overlaid onto a minimal glucose plate (1.5% agar, Vogel-Bonner medium E, 2% glucose). After the overlay solidified, the plates were inverted and placed in an incubator set to maintain 37°C for 67 hours and 29 minutes.

4.10. Experimental Design

Text Table 2
Plate Incorporation Assay

Dose No.	Formulation Conc. (µg/mL)	Dose Volume (µL/plate)	Final Conc. (µg/plate)	Number of Replicates		Number of Strains
				0S9	+S9	
Negative Control	-	100	-	3	3	5
1/ SM-102	19.0 ^b	83.2	1.58	3	3	5
2/ SM-102	58.2 ^b	85.9	5.0	3	3	5
3/ SM-102	184 ^b	85.9	15.8	3	3	5
4/ SM-102	500 ^a	100	50	3	3	5
5/ SM-102	1581 ^a	100	158	3	3	5
6/ SM-102	5000 ^a	100	500	3	3	5
7/ SM-102	15811 ^a	100	1581	3	3	5
8/ SM-102	50000 ^a	100	5000 ^c	3	3	5
Positive controls	d	100	d	3	3	5

- a Theoretical concentrations; actual concentrations may differ slightly due to the limitations of the instruments used.
- b Measured concentration
- c Test Item was tested at levels up to 5000 µg/plate, which is the standard limit dose recommended by regulatory guidelines.
- d Dose depends on the test organism, the positive controls and methodology used (see [Section 4.5.1.2](#))

5. COMPUTERIZED SYSTEMS

Critical computerized systems used in the study are listed below or presented in the appropriate Phase Report. All computerized systems used in the conduct of this study have been validated; when a particular system has not satisfied all requirements, appropriate administrative and procedural controls were implemented to assure the quality and integrity of data.

Text Table 3
Critical Computerized Systems

System Name	Version No.	Description of Data Collected and/or Analyzed
Sorcerer (Perceptive Instruments)	2.2	Revertant colony counts
Microsoft® Excel	2007	Descriptive statistics of revertant colony counts data.
Mesa Laboratories AmegaView CMS	v3.0 Build 1208.8	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO ₂ , as appropriate.
Johnson Controls Metasys	MVE 5.4 (M5)	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories.

6. EVALUATION AND INTERPRETATION OF RESULTS

After the incubation period, the plates from bacterial reverse mutation test were examined visually; an inverted microscope was also employed to facilitate observations. Plates were evaluated for the quality of the background lawn and the number of revertant colonies. Revertant colony counts were collected and saved directly into an electronic database using an automated colony counter. Colony numbers were enumerated visually if precipitation or other artifacts interfered with the colony counter or at the discretion of the Study Director. The presence of visible precipitate was manually recorded in the raw data. Visual counts were performed for this study.

Toxic effects of the Test Item are normally indicated by the partial or complete absence of a background lawn (colony counts, if any, are not reported) or a substantial dose-related reduction in revertant colony counts compared with lower dose levels and concurrent Negative Control (*i.e.* fold response < 0.6) taking into account the laboratory historical control range.

The mean number of revertant colonies for all treatment groups was compared with those obtained for the concurrent Negative Control level.

6.1. Assay Acceptance Criteria:

Acceptable Negative Control: The number of revertants must be within or close the current historical negative control range of the laboratory.

Acceptable Positive Control: The mean value of a Positive Control for a tester strain must be compatible with those generated in the historical positive control data base and be substantially higher (≥ 2 -fold) than the mean number of revertant colonies for its respective Negative Control.

Acceptable highest dose: Depending on the nature of the Test Item, it should reach the toxicity limit, but not exceed 5000 µg/plate, with at least 5 analyzable concentrations, if available.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director was allowed discretion to accept the results of the experiment as valid based on the biological significance.

6.2. Interpretation of Results

Biological relevance of the results was considered first.

Negative result (no evidence of genotoxicity) is concluded if there is no substantial increase (< 2 -fold) in the number of colonies per plate in comparison to the concurrent Negative Control and the data is within or close to the 98% tolerance limit of the Negative Historical Control data. A negative result indicates that the Test Item is non-mutagenic in *S. typhimurium* and *E. coli*.

Positive result (evidence of genotoxicity) is concluded if there is a substantial increase (≥ 2 -fold) in the number of colonies per plate in comparison to the concurrent Negative Control, the mean values are above the 98% tolerance limit of the Negative Historical Control data and a concentration-related increase over the exposure range tested is obtained. A positive result indicates that the Test Item induces point mutations in *S. typhimurium* and/or *E. coli*.

Equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the bacterial reverse mutation assay under the conditions described in the Study Plan. Alternate testing conditions (e.g., a narrower dose interval with the appropriate strain) may be used as an aid in evaluating the test results.

7. RETENTION OF RECORDS

All study-specific raw data, documentation, Study Plan, and final report from this study were archived at the Test Facility by no later than the date of final report issue. One year after issue of the unaudited Draft Report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

Electronic data generated by the Test Facility were archived as noted above, except the reporting files stored on SDMS, which were archived at the Charles River Laboratories facility location in Wilmington, MA.

8. RESULTS

8.1. Dose Formulation Analyses

Results of the dose formulation analyses are presented in [Appendix 4](#).

The five highest formulation concentrations ranging from 500 to 50000 µg/mL met acceptance criteria, with chemical analysis indicating mean achieved concentrations within $\pm 10\%$ of the theoretical concentration for the stock solution and $\pm 15\%$ for lower level solutions. The lowest three formulation concentrations did not meet acceptance criteria (dose number 1 was +20% of nominal, dose number 2 was +16% of nominal and dose number 3 was +16% of nominal). Analyses of the retention samples were not performed due to the consistency of the results of the duplicate samples. Instead the results were accepted and the dose volumes of the three lowest test item concentrations to be tested were adjusted accordingly to obtain the concentrations specified in the study plan.

8.2. Bacterial Mutation Test

Results of the bacterial mutation test are presented in [Table 1](#), [Table 2](#) and [Table 3](#); the mean and standard deviation values quoted have been rounded to the nearest whole number.

The absence of colonies on sterility check plate confirmed absence of microbial contamination (results not shown). The mean revertant colony counts for the Negative Control were within the laboratory historical control range (see [Appendix 3](#) for individual colony count historical results). Appropriate positive controls (with S9 mix where required) induced increases in revertant colony numbers to at least twice the concurrent Negative Control levels with the appropriate bacterial strain, confirming sensitivity of the test system and activity of the S9 mix (see also [Appendix 3](#) for positive controls historical results).

Incomplete, or absent, background lawns of non-revertant bacteria, or substantial reductions in revertant colony counts, were not obtained following exposure to SM-102, indicating that the Test Item was non-toxic to the bacteria at the levels tested. Precipitation was observed at concentrations ≥ 1581 µg/plate in the absence of S9 mix and at concentrations ≥ 500 µg/plate in the presence of S9 mix.

No substantial increases in revertant colony numbers were obtained with any of the tester strains, following exposure to SM-102 at any dose level, in either the presence or absence of S9 mix. Therefore, SM-102 was considered to be negative for the induction of mutagenicity in this in vitro assay.

9. CONCLUSION

It is concluded that SM-102 did not show any evidence of genotoxic activity in this in vitro mutagenicity assay when tested in accordance with regulatory guidelines.

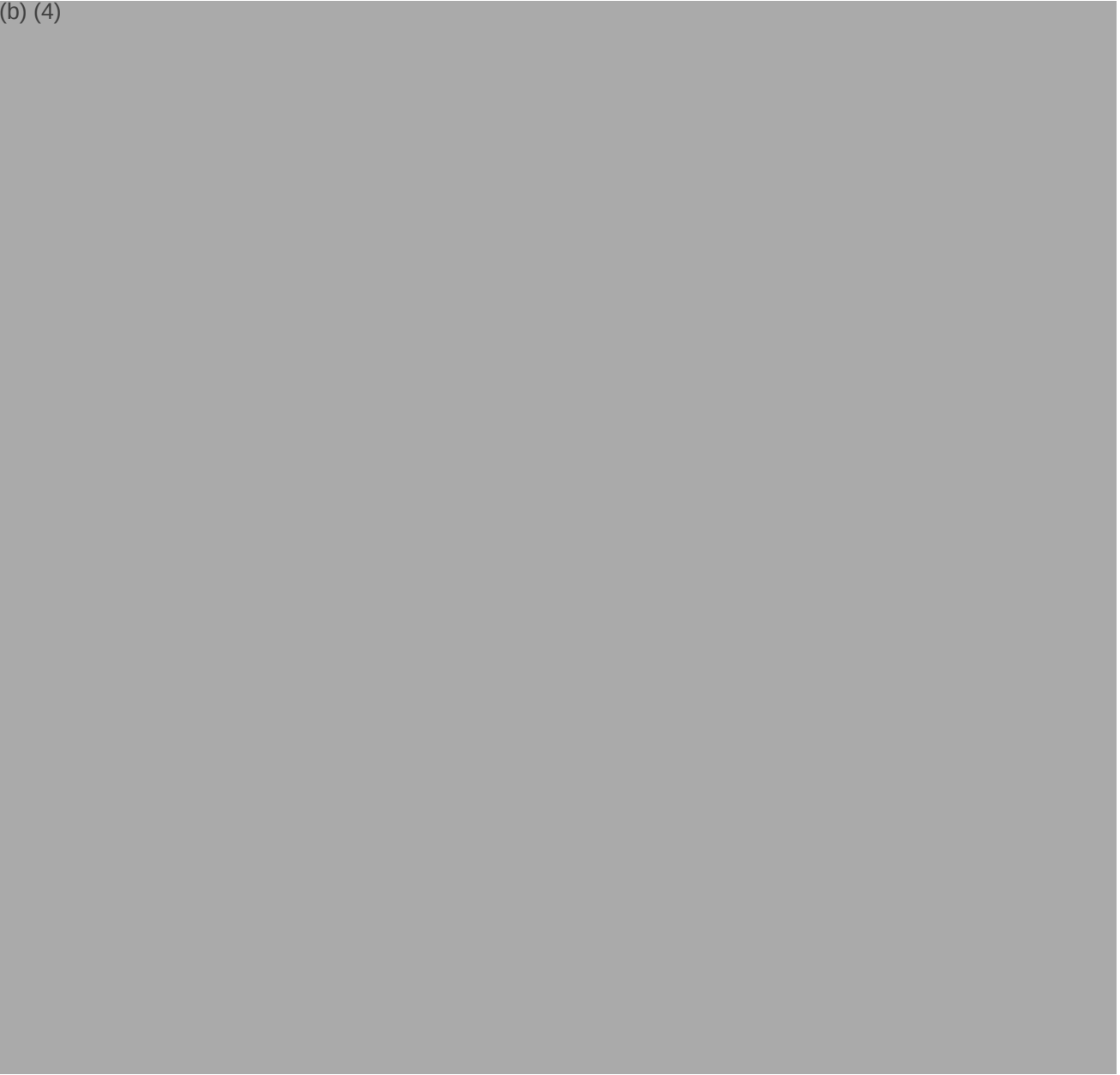
(b) (4)



(b) (4)



(b) (4)



(b) (4)



(b) (4)



Appendix 1
Study Plan, Amendment and Deviations



FINAL STUDY PLAN

Test Facility Study No. 9601567

SM-102

**Bacterial Reverse Mutation Test in *Salmonella typhimurium* and
*Escherichia coli***

SPONSOR:

Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139, USA

TEST FACILITY:

Charles River Laboratories Montreal ULC

(b) (6)

A large rectangular area of the document is redacted with a solid grey fill, covering several lines of text.

08 September 2016

Page 1 of 14

TABLE OF CONTENTS

1. OBJECTIVE3

2. PROPOSED STUDY SCHEDULE3

3. GUIDELINES FOR STUDY DESIGN3

4. REGULATORY COMPLIANCE3

5. QUALITY ASSURANCE4

6. SPONSOR4

7. RESPONSIBLE PERSONNEL4

8. TEST AND REFERENCE ITEMS5

9. SAFETY6

10. DOSE FORMULATION AND ANALYSIS6

11. TEST SYSTEM7

12. EXPERIMENTAL DESIGN8

13. EXPERIMENTAL PROCEDURES9

14. EVALUATION AND INTERPRETATION OF RESULTS9

15. COMPUTERIZED SYSTEMS11

16. AMENDMENTS AND DEVIATIONS11

17. RETENTION OF RECORDS11

18. REPORTING12

TEST FACILITY APPROVAL13

SPONSOR APPROVAL14

1. OBJECTIVE

The objective of this study is to determine the potential genotoxicity of SM-102, using the bacterial reverse mutation test.

2. PROPOSED STUDY SCHEDULE

Proposed study dates are listed below. Actual applicable dates will be included in the Final Report.

Experimental Start Date:	12 Sep 2016 (First date of study-specific data collection)
Experimental Completion Date:	21 Sep 2016 (Last date data are collected from the study)
Initiation of Dosing:	13 Sep 2016
Unaudited Draft Report:	12 Oct 2016
Final Report	See Reporting Section 18

3. GUIDELINES FOR STUDY DESIGN

The design of this study was based on the study objective, the overall product development strategy for the Test Item, and the following study design guidelines:

- OECD Guideline 471. *OECD Guideline for Testing of Chemicals - Bacterial Reverse Mutation Test.*
- ICH Harmonised Tripartite Guideline S2(R1). *Guidance on Genotoxicity Testing and Data Interpretation for Pharmaceuticals Intended for Human Use.*

4. REGULATORY COMPLIANCE

The study will be performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA) and Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions to GLPs include the following study elements:

- Characterization of the Test Item will be performed by the Sponsor or Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses will not be conducted in compliance with the GLP or GMP regulations.
- Stability testing of the supplied Test Item will be performed by the Sponsor or Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

5. QUALITY ASSURANCE

5.1. Test Facility

The Test Facility Quality Assurance Program (QAP) will monitor the study to assure the facilities, equipment, personnel, methods, practices, records, and controls are in conformance with Good Laboratory Practice regulations. The QAP will review the Study Plan, conduct inspections at intervals adequate to assure the integrity of the study, and audit the Final Report to assure that it accurately describes the methods and standard operating procedures and that the reported results accurately reflect the raw data of the study.

6. SPONSOR

Sponsor Representative

(b) (6)
Address as cited for the Sponsor
Tel: (b) (6)
E-mail: (b) (6)

7. RESPONSIBLE PERSONNEL

Study Director

(b) (6)
Address as cited for Test Facility
Tel: (b) (6)
Fax: (b) (6)
E-mail: (b) (6)

Management Contact

(b) (6)
Address as cited for Test Facility
Tel: (b) (6)
Fax: (b) (6)
E-mail: (b) (6)

Individual Scientist (IS) at the Test Facility

Analytical Chemistry (b) (6)
Address as cited for Test Facility
Tel: (b) (6)
E-mail: (b) (6)

Each IS is required to report any deviations or other circumstances that could affect the quality or integrity of the study to the Study Director in a timely manner. Each IS will provide a report addressing their assigned phase of the study, which will be included as an appendix to the Final Report. The phase report will include the following:

- A listing of critical computerized systems used in the conduct and/or interpretation of the assigned study phase

8. TEST AND REFERENCE ITEMS

8.1. Test Item

Identification: SM-102
Batch (Lot) No.: RL-100-211-1
Retest Date: To be included in the final report
Purity: 95.3%; dose calculations will be corrected for purity
Storage Conditions: Kept in a freezer set to maintain -20°C

8.2. Reference Items

8.2.1. Negative Control

Identification: Ethanol

Full details of the negative control, including supplier, lot number, storage, expiry date, will be maintained with the study raw data; appropriate details will be presented in the Final Report.

8.2.2. Positive Controls

In the Absence of S9 Mix:

Sodium azide, 9-aminoacridine hemihydrate, 2-nitrofluorene, 4-nitroquinoline N-oxide

In the Presence of S9 Mix:

2-Aminoanthracene and benzo[a]pyrene

Full details of the positive controls, including supplier, lot number, storage, expiry date, formulation methods and concentrations (if applicable), will be maintained with the study raw data or as part of the Test Facility records; appropriate details will be presented in the final report.

8.3. Test Item Characterization

The Sponsor will provide to the Test Facility documentation of the identity, strength, purity, composition, and stability for the Test Item. A Certificate of Analysis or equivalent documentation will be provided for inclusion in the Final Report. The Sponsor will also provide information concerning the regulatory standard that was followed for these evaluations.

The Sponsor has appropriate documentation on file concerning the method of synthesis, fabrication or derivation of the Test Item, and this information is available to the appropriate regulatory agencies should it be requested.

8.4. Analysis of Test Item

The stability of the bulk Test Item will not be determined during the course of this study. Information to support the stability of each lot of bulk Test Item will be provided by the Sponsor.

8.5. Test Item Inventory and Disposition

Records of the receipt, distribution, storage and disposition of Test Item will be maintained. All unused Sponsor-supplied bulk Test Item will be returned to the Sponsor or discarded as instructed by the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

9. SAFETY

The safety precautions for the Test Item and dose formulations will be based on the information provided by the Sponsor either by an MSDS or similar document.

10. DOSE FORMULATION AND ANALYSIS

10.1. Preparation of Reference Items

10.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, will be dispensed into a vial for administration to control plates. The aliquot will be stored in a refrigerator set to maintain 4°C until use. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.1.2. Preparation of Positive Controls

The Positive Control formulations will be prepared up to 6 months prior to use or on the day of use. The Positive Controls will be dissolved in the appropriate vehicle at the selected concentration. Adequate amounts of the formulations will be dispensed into vials, and stored in a freezer set to maintain -20°C, protected from light, until use. The aliquots will be removed from the freezer and allowed to warm to ambient room temperature before dosing. Any residual volumes will be discarded after completion of dosing.

Depending on the strains tested, the following Positive Controls will be used:

Positive Controls for the Assay

Strain	S9	Positive Controls	Concentration µg/plate	Vehicle
TA1535, TA100	0	Sodium azide (NaAz)	0.5	Sterile water
TA1537		9-Aminoacridine hemihydrate (9AC)	50	DMSO
TA98		2-Nitrofluorene (2NF)	1	DMSO
WP2 <i>uvrA</i>		4-Nitroquinoline N-oxide (NQO)	0.5	DMSO
TA1535	+	2-Aminoanthracene (2AA)	5	DMSO
WP2 <i>uvrA</i>			20	
TA1537, TA98, TA100		Benzo[a]pyrene (BaP)	5	DMSO

10.2. Preparation of Test Item

Test Item dosing formulations will be prepared at appropriate concentrations to meet dosage level requirements. The Test Item will be prepared as a stock solution (50 mg/mL) in the vehicle (ethanol) and all lower level formulations will be prepared by serial dilution. The formulations will be prepared up to 4 days prior to use and will be stored in a refrigerator set to maintain 4°C

until use. The formulations will be removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.3. Sample Collection and Analysis

Dose formulation samples will be collected from the dosing vial for analysis as indicated in the following table. Additional samples may be collected and analyzed at the discretion of the Study Director. The Positive Control formulations will not be subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Dose Formulation Sample Collection Schedule

Concentration
Dose numbers 1 to 8

All samples to be analyzed will be transferred under ambient conditions to the analytical laboratory at the Test Facility. Any residual/retained analytical samples will be discarded before issuance of the Final Report.

10.3.1. Analytical Method

Analyses described below will be performed by HPLC using a validated analytical procedure (Test Facility under Study Number 1801841).

10.3.1.1. Concentration Analysis

Samples for Analysis:	Duplicate samples, sent for analysis as noted in Section 10.3
Backup Samples:	Duplicate samples, maintained at the Test Facility. Backup samples may be analyzed at the discretion of the Study Director.
Sample Volume:	1 mL for dose numbers 1 and 2, 0.5 mL for dose numbers 3 to 8
Sampling Containers:	Appropriate sized glass containers
Storage Conditions:	Kept in a refrigerator set to maintain 4°C
Acceptance Criteria:	Chemically analyzed mean values are normally expected to be within $\pm 10\%$ of theoretical concentration for stock solutions and $\pm 15\%$ for lower level solutions. The potential impact of any departure outside these limits will be addressed in the report.

10.3.1.2. Stability Analysis

Stability analyses performed previously at the Test Facility under Study Number 1801841 demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study. Stability data have been retained in the study records for Study No. 1801841.

11. TEST SYSTEM

The bacteria were originally supplied by Moltox, NC, USA. Each bacterial culture is identified by its name (TA1535, TA1537, TA98, TA100, WP2 *uvrA*) and batch number. Each batch of

frozen bacteria is tested for appropriate phenotype characteristics, spontaneous reversion rate and response to diagnostic mutagens. The following bacterial strains will be employed:

- S. typhimurium* TA1535 *hisG46 rfa ΔuvrB*
- S. typhimurium* TA1537 *hisC3076 rfa ΔuvrB*
- S. typhimurium* TA98 *hisD3052 rfa ΔuvrB* pKM101
- S. typhimurium* TA100 *hisG46 rfa ΔuvrB* pKM101
- E. coli* WP2 *trp uvrA*

Fresh bacterial cultures will be prepared and will be in the late-log phase of growth at the time of use. The density of the cultures will be confirmed to be $\geq 1000 \times 10^6$ bacteria/mL using a bacterial counting chamber before the cultures are used in the test.

11.1. Reason for Choice of Test System

The bacterial mutation test detects point mutations, which are the cause of many human genetic diseases and can play a role in tumor initiation and development.

11.2. Reason for Dose Level Selection

Typically, the Test Item is dosed at a range of concentrations but is only assessed at the five highest levels below the toxic level or, if non-toxic, at five levels up to the standard limit of 5000 µg/plate.

12. EXPERIMENTAL DESIGN

Experimental Design Plate Incorporation Assay

Dose No.	Formulation Conc. (µg/mL) ^a	Final Conc. (µg/plate)	Number of Replicates		Number of Strains
			0S9	+S9	
Negative Control	-	-	3	3	5
1/ SM-102	15.8	1.58	3	3	5
2/ SM-102	50	5.0	3	3	5
3/ SM-102	158	15.8	3	3	5
4/ SM-102	500	50	3	3	5
5/ SM-102	1581	158	3	3	5
6/ SM-102	5000	500	3	3	5
7/ SM-102	15811	1581	3	3	5
8/ SM-102	50000	5000 ^b	3	3	5
Positive controls	c	c	3	3	5

a Theoretical concentrations; actual concentrations may be adjusted due to the limitations of the instruments used.

b The OECD/ICH S2(R1) standard limit dose (usually 5000 µg/plate). Equivocal findings, excessive toxicity or missing/invalid results may require supplemental testing using appropriate strain(s) and S9 conditions over an appropriate dose range (to be added by Study Plan amendment).

c Dose depends on the test organism, the positive controls and methodology used (see Section 10.1.2)

13. EXPERIMENTAL PROCEDURES

13.1. S9 Mix

The S9 mix, used as a model of intact mammalian metabolism, will be prepared on the day of use and will contain 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM MgCl₂, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate and 4 mM NADP. The S9 mix will be stored in a refrigerator set to maintain 4°C until required and then held on ice during utilization. A copy of the manufacturer's quality control certificate(s) for the S9 fraction will be retained as raw data.

13.2. Sterility Check and Spontaneous Mutation Rates

The sterility of the Test Item formulation (high dose) will be assessed on the day of the test using appropriate preparations without bacteria.

The spontaneous mutation rates of the bacterial strains will be assessed using concurrent control samples in which the bacteria are exposed to the Negative Control.

13.3. Plate Incorporation Method

A 0.5 mL aliquot of S9 mix (+S9) or phosphate buffer 0.2M pH 7.4 (0S9) will be combined with 0.1 mL bacterial culture in a sterile container. A standard volume (typically 0.1 mL) of the test/reference items will be added, and approximately 2 mL of molten top agar supplemented with 0.05 mM biotin and minimal (0.05 mM) histidine and minimal (0.05 mM) tryptophan will be added immediately afterward. The solution will be mixed and overlaid onto a minimal glucose plate (1.5% agar, Vogel-Bonner medium E, 2% glucose). After the overlay solidifies, the plates will be inverted and place in an incubator set to maintain 37°C for 60 to 72 h. The sterile containers and the plates will be labeled with the study number and the number that refers to the experimental design.

14. EVALUATION AND INTERPRETATION OF RESULTS

After the incubation period, the plates will be examined visually and with the aid of an inverted microscope. Plates will be evaluated for the quality of the background lawn and the number of revertant colonies. Revertant colony counts are routinely collected and saved directly into an electronic database using an automated colony counter. Colony numbers will be enumerated visually if precipitation or other artifacts interfere with the colony counter (if practical), or at the Study Director's discretion. Any visual counts and the presence of visible precipitate are manually recorded in the raw data. If precipitation interferes with observation of the background lawn, this is recorded and noted in the report. In such cases, the background lawn is assumed to be normal if the colony counts are similar to the Negative Control. In the rare event that precipitation prevents counting of revertant colonies, this is noted in the report and the colony count is considered not readable.

If available, the plates from at least five non-toxic dose levels of the Test Item will be assessed, i.e. the five highest levels below the toxic level. If the Test Item is non-toxic, then the five levels up to the standard limit of 5000 µg/plate will be assessed. Toxic effects of the Test Item are normally indicated by the partial or complete absence of a background lawn (colony counts, if

any, will not be reported) or a substantial dose-related reduction in revertant colony counts compared with lower dose levels and concurrent negative control (i.e., fold response < 0.6) taking into account the laboratory historical control range.

Five or more non-toxic dose levels are preferred for evaluating the study. Four non-toxic concentrations are permitted to evaluate the study. In the event that less than 4 non-toxic dose levels are achieved in the study, the results will be evaluated with the Sponsor and it will be decided if the affected portion of the study will be repeated with an appropriate change in dose levels. If a repeat is required, this will be documented by a study plan amendment.

Means and standard deviations for appropriate dose levels will be calculated and presented in the report together with laboratory historical control information. The mean number of revertant colonies for all treatment groups will be compared with those obtained for the concurrent negative control level.

Colony counts and mean colony counts will generally be reported to the nearest integer value. Where a validated data acquisition and reporting system is in use, the reporting format of that system will be used. Fold changes against control values will generally be tabulated to 1 decimal place. Fold increases against control may occasionally be reported to 2 decimal places to clarify the conclusion with respect to evaluation criteria.

14.1. Assay acceptance criteria:

Acceptable Negative Control: The number of revertants must be within or close the current historical negative control range of the laboratory.

Acceptable Positive Control: The mean value of a Positive Control for a tester strain must be compatible with those generated in the historical positive control data base and be substantially higher (≥ 2 -fold) than the mean number of revertant colonies for its respective Negative Control.

Acceptable highest dose: Depending on the nature of the Test Item, it should reach the toxicity limit (as per Section 11.2), but not exceed 5000 $\mu\text{g}/\text{plate}$, with at least 5 analyzable concentrations, if available.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director will be allowed discretion to accept the results of the experiment as valid based on the biological significance.

14.2. Interpretation of Results

Biological relevance of the results will be considered first.

Negative result (no evidence of genotoxicity) is concluded if there is no substantial increase (< 2 -fold) in the number of colonies per plate in comparison to the concurrent Negative Control and the data is within or close to the 98% tolerance limit of the Negative Historical Control data. A negative result indicates that the Test Item is non-mutagenic in *S. typhimurium* and *E. coli*.

Positive result (evidence of genotoxicity) is concluded if there is a substantial increase (≥ 2 -fold) in the number of colonies per plate in comparison to the concurrent Negative Control, the mean values are above the 98% tolerance limit of the Negative Historical Control data and a

concentration-related increase over the exposure range tested is obtained. A positive result indicates that the Test Item induces point mutations in *S. typhimurium* and/or *E. coli*.

Equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the bacterial reverse mutation assay under the conditions described in this study plan. Alternate testing conditions (e.g., a narrower dose interval with the appropriate strain) may be used as an aid in evaluating the test results. Any additional testing or analysis will be approved by the sponsor and will be documented by study plan amendment.

15. COMPUTERIZED SYSTEMS

The following proposed critical computerized systems may be used in the study. The actual critical computerized systems used will be specified in the Final Report.

Data for parameters not required by Study Plan, which are automatically generated by analytical devices used will be retained on file but not reported. Statistical analysis results that are generated by the program but are not required by Study Plan and/or are not scientifically relevant will be retained on file but will not be included in the tabulations.

Proposed Critical Computerized Systems

System Name	Description of Data Collected and/or Analyzed
Sorcerer (Perceptive Instruments)	Revertant colony counts.
Excel (Microsoft)	Descriptive statistics of revertant colony counts data.
Mesa Laboratories AmegaView CMS	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO ₂ , as appropriate.
Johnson Controls Metasys	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories.
Empower 3 (Waters Corporation)	Dose formulation analyses using HPLC.
Empower 3 and Microsoft Excel	Regression analysis and descriptive statistics for dose formulation analytical data.

16. AMENDMENTS AND DEVIATIONS

Changes to the approved Study Plan shall be made in the form of an amendment, which will be signed and dated by the Study Director. Every reasonable effort will be made to discuss any necessary Study Plan changes in advance with the Sponsor.

All Study Plan and SOP deviations will be documented in the study records. The Study Director will notify the Sponsor of deviations that may result in a significant impact on the study as soon as possible.

17. RETENTION OF RECORDS

All study-specific raw data, electronic data, documentation, study plan, retained samples, and interim (if applicable) and final reports will be archived by no later than the date of final report issue. All materials generated by Charles River from this study will be transferred to a Charles

River archive. One year after issue of the unaudited draft report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

18. REPORTING

A comprehensive unaudited Draft Report will be prepared following completion of the study and will be finalized following consultation with the Sponsor. The report will include all information necessary to provide a complete and accurate description of the experimental methods and results and any circumstances that may have affected the quality or integrity of the study.

The Sponsor will receive an electronic version of the unaudited Draft Report provided in Microsoft Word format and an electronic version of the Final Report provided in Adobe Acrobat PDF format (hyperlinked and searchable). The PDF document will be created from native electronic files to the every possible extent, including text and tables generated by the Test Facility. Report components not available in native electronic files and/or original signature pages will be scanned and converted to PDF image files for incorporation.

Reports should be finalized within 6 months of issuance of the unaudited Draft Report. If the Sponsor has not provided comments to the report within 6 months of draft issuance, the report will be finalized by the Test Facility unless other arrangements are made by the Sponsor.

TEST FACILITY APPROVAL

The signature below acknowledges Test Facility Management's responsibility to the study as defined by the relevant GLP regulations.

(b) (6)

Date: 08 Sep 2016

The signature below indicates that the Study Director approves the Study Plan.

(b) (6)

Date: 08 Sep 2016

SPONSOR APPROVAL

The Study Plan was approved by the Sponsor by email on 07 Sep 2016. The signature below confirms the approval of the Study Plan by the Sponsor Representative.

(b) (6)

Date: 15-Sep-16



STUDY PLAN AMENDMENT NO. 1

Test Facility Study No. 9601567

SM-102

**Bacterial Reverse Mutation Test in *Salmonella typhimurium* and
*Escherichia coli***

SPONSOR:

Moderna Therapeutics, Inc.
200 Technology Square, Third Floor
Cambridge, MA 02139, USA

TEST FACILITY:

Charles River Laboratories Montreal ULC

(b) (6)

A large rectangular area of the document is redacted with a solid grey fill, obscuring several lines of text.

13 September 2016

Page 1 of 14

SUMMARY OF CHANGES AND JUSTIFICATIONS

Study Plan effective date: Date: 08 Sep 2016

Note: When applicable, additions are indicated in bold underlined text and deletions are indicated in bold strikethrough text in the affected sections of the document.

Section	Justification
Amendment 1	Date: 13 Sep 2016
12. Experimental Design	Dose volume changed for dose numbers 1, 2 and 3 after results from the dose formulation analysis indicated that the formulation concentrations were above the acceptance criteria.

TABLE OF CONTENTS

SUMMARY OF CHANGES AND JUSTIFICATIONS	2
1. OBJECTIVE.....	4
2. PROPOSED STUDY SCHEDULE.....	4
3. GUIDELINES FOR STUDY DESIGN	4
4. REGULATORY COMPLIANCE	4
5. QUALITY ASSURANCE	5
6. SPONSOR.....	5
7. RESPONSIBLE PERSONNEL.....	5
8. TEST AND REFERENCE ITEMS	6
9. SAFETY	7
10. DOSE FORMULATION AND ANALYSIS	7
11. TEST SYSTEM.....	8
12. EXPERIMENTAL DESIGN.....	9
13. EXPERIMENTAL PROCEDURES.....	10
14. EVALUATION AND INTERPRETATION OF RESULTS	10
15. COMPUTERIZED SYSTEMS	12
16. AMENDMENTS AND DEVIATIONS.....	12
17. RETENTION OF RECORDS	12
18. REPORTING.....	13
AMENDMENT APPROVAL	14

1. OBJECTIVE

The objective of this study is to determine the potential genotoxicity of SM-102, using the bacterial reverse mutation test.

2. PROPOSED STUDY SCHEDULE

Proposed study dates are listed below. Actual applicable dates will be included in the Final Report.

Experimental Start Date:	12 Sep 2016 (First date of study-specific data collection)
Experimental Completion Date:	21 Sep 2016 (Last date data are collected from the study)
Initiation of Dosing:	13 Sep 2016
Unaudited Draft Report:	12 Oct 2016
Final Report	See Reporting Section 18

3. GUIDELINES FOR STUDY DESIGN

The design of this study was based on the study objective, the overall product development strategy for the Test Item, and the following study design guidelines:

- OECD Guideline 471. *OECD Guideline for Testing of Chemicals - Bacterial Reverse Mutation Test.*
- ICH Harmonised Tripartite Guideline S2(R1). *Guidance on Genotoxicity Testing and Data Interpretation for Pharmaceuticals Intended for Human Use.*

4. REGULATORY COMPLIANCE

The study will be performed in accordance with the OECD Principles of Good Laboratory Practice and as accepted by Regulatory Authorities throughout the European Union, United States of America (FDA) and Japan (MHLW), and other countries that are signatories to the OECD Mutual Acceptance of Data Agreement.

Exceptions to GLPs include the following study elements:

- Characterization of the Test Item will be performed by the Sponsor or Sponsor subcontractor according to established SOPs, controls and approved test methodologies to ensure integrity and validity of the results generated; these analyses will not be conducted in compliance with the GLP or GMP regulations.
- Stability testing of the supplied Test Item will be performed by the Sponsor or Sponsor subcontractor at a laboratory that follows FDA Good Manufacturing Practice (GMP) regulations.

5. QUALITY ASSURANCE

5.1. Test Facility

The Test Facility Quality Assurance Program (QAP) will monitor the study to assure the facilities, equipment, personnel, methods, practices, records, and controls are in conformance with Good Laboratory Practice regulations. The QAP will review the Study Plan, conduct inspections at intervals adequate to assure the integrity of the study, and audit the Final Report to assure that it accurately describes the methods and standard operating procedures and that the reported results accurately reflect the raw data of the study.

6. SPONSOR

Sponsor Representative

(b) (6)
Address as cited for the Sponsor
Tel: (b) (6)
E-mail: (b) (6)

7. RESPONSIBLE PERSONNEL

Study Director

(b) (6)
Address as cited for Test Facility
Tel: (b) (6)
Fax: (b) (6)
E-mail: (b) (6)

Management Contact

(b) (6)
Address as cited for Test Facility
Tel: (b) (6)
Fax: (b) (6)
E-mail: (b) (6)

Individual Scientist (IS) at the Test Facility

Analytical Chemistry (b) (6)
Address as cited for Test Facility
Tel: (b) (6)
E-mail: (b) (6)

Each IS is required to report any deviations or other circumstances that could affect the quality or integrity of the study to the Study Director in a timely manner. Each IS will provide a report addressing their assigned phase of the study, which will be included as an appendix to the Final Report. The phase report will include the following:

- A listing of critical computerized systems used in the conduct and/or interpretation of the assigned study phase

8. TEST AND REFERENCE ITEMS

8.1. Test Item

Identification: SM-102
Batch (Lot) No.: RL-100-211-1
Retest Date: To be included in the final report
Purity: 95.3%; dose calculations will be corrected for purity
Storage Conditions: Kept in a freezer set to maintain -20°C

8.2. Reference Items

8.2.1. Negative Control

Identification: Ethanol

Full details of the negative control, including supplier, lot number, storage, expiry date, will be maintained with the study raw data; appropriate details will be presented in the Final Report.

8.2.2. Positive Controls

In the Absence of S9 Mix:

Sodium azide, 9-aminoacridine hemihydrate, 2-nitrofluorene, 4-nitroquinoline N-oxide

In the Presence of S9 Mix:

2-Aminoanthracene and benzo[a]pyrene

Full details of the positive controls, including supplier, lot number, storage, expiry date, formulation methods and concentrations (if applicable), will be maintained with the study raw data or as part of the Test Facility records; appropriate details will be presented in the final report.

8.3. Test Item Characterization

The Sponsor will provide to the Test Facility documentation of the identity, strength, purity, composition, and stability for the Test Item. A Certificate of Analysis or equivalent documentation will be provided for inclusion in the Final Report. The Sponsor will also provide information concerning the regulatory standard that was followed for these evaluations.

The Sponsor has appropriate documentation on file concerning the method of synthesis, fabrication or derivation of the Test Item, and this information is available to the appropriate regulatory agencies should it be requested.

8.4. Analysis of Test Item

The stability of the bulk Test Item will not be determined during the course of this study. Information to support the stability of each lot of bulk Test Item will be provided by the Sponsor.

8.5. Test Item Inventory and Disposition

Records of the receipt, distribution, storage and disposition of Test Item will be maintained. All unused Sponsor-supplied bulk Test Item will be returned to the Sponsor or discarded as instructed by the Sponsor following completion of the experimental phase of the study. Any remaining Reference Items (negative and positive controls) will be retained at the Test Facility or discarded upon expiry.

9. SAFETY

The safety precautions for the Test Item and dose formulations will be based on the information provided by the Sponsor either by an MSDS or similar document.

10. DOSE FORMULATION AND ANALYSIS

10.1. Preparation of Reference Items

10.1.1. Preparation of Negative Control

An adequate amount of the Negative Control, ethanol, will be dispensed into a vial for administration to control plates. The aliquot will be stored in a refrigerator set to maintain 4°C until use. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.1.2. Preparation of Positive Controls

The Positive Control formulations will be prepared up to 6 months prior to use or on the day of use. The Positive Controls will be dissolved in the appropriate vehicle at the selected concentration. Adequate amounts of the formulations will be dispensed into vials, and stored in a freezer set to maintain -20°C, protected from light, until use. The aliquots will be removed from the freezer and allowed to warm to ambient room temperature before dosing. Any residual volumes will be discarded after completion of dosing.

Depending on the strains tested, the following Positive Controls will be used:

Positive Controls for the Assay

Strain	S9	Positive Controls	Concentration µg/plate	Vehicle
TA1535, TA100	0	Sodium azide (NaAz)	0.5	Sterile water
TA1537		9-Aminoacridine hemihydrate (9AC)	50	DMSO
TA98		2-Nitrofluorene (2NF)	1	DMSO
WP2 <i>uvrA</i>		4-Nitroquinoline N-oxide (NQO)	0.5	DMSO
TA1535	+	2-Aminoanthracene (2AA)	5	DMSO
WP2 <i>uvrA</i>			20	
TA1537, TA98, TA100		Benzo[a]pyrene (BaP)	5	DMSO

10.2. Preparation of Test Item

Test Item dosing formulations will be prepared at appropriate concentrations to meet dosage level requirements. The Test Item will be prepared as a stock solution (50 mg/mL) in the vehicle (ethanol) and all lower level formulations will be prepared by serial dilution. The formulations will be prepared up to 4 days prior to use and will be stored in a refrigerator set to maintain 4°C

until use. The formulations will be removed from the refrigerator and allowed to warm to room temperature for at least 30 minutes before dosing. Any residual volumes will be discarded unless otherwise requested by the Study Director.

10.3. Sample Collection and Analysis

Dose formulation samples will be collected from the dosing vial for analysis as indicated in the following table. Additional samples may be collected and analyzed at the discretion of the Study Director. The Positive Control formulations will not be subjected to analysis for safety reasons and because the biological response of the test system is considered to be the best measure of the appropriateness of the formulations.

Dose Formulation Sample Collection Schedule

Concentration
Dose numbers 1 to 8

All samples to be analyzed will be transferred under ambient conditions to the analytical laboratory at the Test Facility. Any residual/retained analytical samples will be discarded before issuance of the Final Report.

10.3.1. Analytical Method

Analyses described below will be performed by HPLC using a validated analytical procedure (Test Facility under Study Number 1801841).

10.3.1.1. Concentration Analysis

Samples for Analysis:	Duplicate samples, sent for analysis as noted in Section 10.3
Backup Samples:	Duplicate samples, maintained at the Test Facility. Backup samples may be analyzed at the discretion of the Study Director.
Sample Volume:	1 mL for dose numbers 1 and 2, 0.5 mL for dose numbers 3 to 8
Sampling Containers:	Appropriate sized glass containers
Storage Conditions:	Kept in a refrigerator set to maintain 4°C
Acceptance Criteria:	Chemically analyzed mean values are normally expected to be within $\pm 10\%$ of theoretical concentration for stock solutions and $\pm 15\%$ for lower level solutions. The potential impact of any departure outside these limits will be addressed in the report.

10.3.1.2. Stability Analysis

Stability analyses performed previously at the Test Facility under Study Number 1801841 demonstrated that the Test Item is stable in the vehicle when prepared and stored under the same conditions at concentrations bracketing those used in the present study. Stability data have been retained in the study records for Study No. 1801841.

11. TEST SYSTEM

The bacteria were originally supplied by Moltox, NC, USA. Each bacterial culture is identified by its name (TA1535, TA1537, TA98, TA100, WP2 *uvrA*) and batch number. Each batch of

frozen bacteria is tested for appropriate phenotype characteristics, spontaneous reversion rate and response to diagnostic mutagens. The following bacterial strains will be employed:

- S. typhimurium* TA1535 *hisG46 rfa ΔuvrB*
- S. typhimurium* TA1537 *hisC3076 rfa ΔuvrB*
- S. typhimurium* TA98 *hisD3052 rfa ΔuvrB* pKM101
- S. typhimurium* TA100 *hisG46 rfa ΔuvrB* pKM101
- E. coli* WP2 *trp uvrA*

Fresh bacterial cultures will be prepared and will be in the late-log phase of growth at the time of use. The density of the cultures will be confirmed to be $\geq 1000 \times 10^6$ bacteria/mL using a bacterial counting chamber before the cultures are used in the test.

11.1. Reason for Choice of Test System

The bacterial mutation test detects point mutations, which are the cause of many human genetic diseases and can play a role in tumor initiation and development.

11.2. Reason for Dose Level Selection

Typically, the Test Item is dosed at a range of concentrations but is only assessed at the five highest levels below the toxic level or, if non-toxic, at five levels up to the standard limit of 5000 µg/plate.

12. EXPERIMENTAL DESIGN

Experimental Design Plate Incorporation Assay

Dose No.	Formulation Conc. (µg/mL)*	Dose Volume (µL)	Final Conc. (µg/plate)	Number of Replicates		Number of Strains
				0S9	+S9	
Negative Control	-	100	-	3	3	5
1/ SM-102	15.8 ^d	83.2	1.58	3	3	5
2/ SM-102	50 ^d	85.9	5.0	3	3	5
3/ SM-102	158 ^d	85.9	15.8	3	3	5
4/ SM-102	500 ^a	100	50	3	3	5
5/ SM-102	1581 ^a	100	158	3	3	5
6/ SM-102	5000 ^a	100	500	3	3	5
7/ SM-102	15811 ^a	100	1581	3	3	5
8/ SM-102	50000 ^a	100	5000 ^b	3	3	5
Positive controls	c	100	c	3	3	5

a Theoretical concentrations; actual concentrations may be adjusted due to the limitations of the instruments used.

b The OECD/ICH S2(R1) standard limit dose (usually 5000 µg/plate). Equivocal findings, excessive toxicity or missing/invalid results may require supplemental testing using appropriate strain(s) and S9 conditions over an appropriate dose range (to be added by Study Plan amendment).

c Dose depends on the test organism, the positive controls and methodology used (see Section 10.1.2)

d Measured concentration

13. EXPERIMENTAL PROCEDURES

13.1. S9 Mix

The S9 mix, used as a model of intact mammalian metabolism, will be prepared on the day of use and will contain 10% v/v S9 fraction (Aroclor 1254 induced male rat liver fraction supplied by Moltox) and the following sterile cofactors: 8 mM MgCl₂, 33 mM KCl, 100 mM sodium phosphate buffer pH 7.4, 5 mM glucose-6-phosphate and 4 mM NADP. The S9 mix will be stored in a refrigerator set to maintain 4°C until required and then held on ice during utilization. A copy of the manufacturer's quality control certificate(s) for the S9 fraction will be retained as raw data.

13.2. Sterility Check and Spontaneous Mutation Rates

The sterility of the Test Item formulation (high dose) will be assessed on the day of the test using appropriate preparations without bacteria.

The spontaneous mutation rates of the bacterial strains will be assessed using concurrent control samples in which the bacteria are exposed to the Negative Control.

13.3. Plate Incorporation Method

A 0.5 mL aliquot of S9 mix (+S9) or phosphate buffer 0.2M pH 7.4 (0S9) will be combined with 0.1 mL bacterial culture in a sterile container. A standard volume (typically 0.1 mL) of the test/reference items will be added, and approximately 2 mL of molten top agar supplemented with 0.05 mM biotin and minimal (0.05 mM) histidine and minimal (0.05 mM) tryptophan will be added immediately afterward. The solution will be mixed and overlaid onto a minimal glucose plate (1.5% agar, Vogel-Bonner medium E, 2% glucose). After the overlay solidifies, the plates will be inverted and place in an incubator set to maintain 37°C for 60 to 72 h. The sterile containers and the plates will be labeled with the study number and the number that refers to the experimental design.

14. EVALUATION AND INTERPRETATION OF RESULTS

After the incubation period, the plates will be examined visually and with the aid of an inverted microscope. Plates will be evaluated for the quality of the background lawn and the number of revertant colonies. Revertant colony counts are routinely collected and saved directly into an electronic database using an automated colony counter. Colony numbers will be enumerated visually if precipitation or other artifacts interfere with the colony counter (if practical), or at the Study Director's discretion. Any visual counts and the presence of visible precipitate are manually recorded in the raw data. If precipitation interferes with observation of the background lawn, this is recorded and noted in the report. In such cases, the background lawn is assumed to be normal if the colony counts are similar to the Negative Control. In the rare event that precipitation prevents counting of revertant colonies, this is noted in the report and the colony count is considered not readable.

If available, the plates from at least five non-toxic dose levels of the Test Item will be assessed, i.e. the five highest levels below the toxic level. If the Test Item is non-toxic, then the five levels up to the standard limit of 5000 µg/plate will be assessed. Toxic effects of the Test Item are normally indicated by the partial or complete absence of a background lawn (colony counts, if

any, will not be reported) or a substantial dose-related reduction in revertant colony counts compared with lower dose levels and concurrent negative control (i.e., fold response < 0.6) taking into account the laboratory historical control range.

Five or more non-toxic dose levels are preferred for evaluating the study. Four non-toxic concentrations are permitted to evaluate the study. In the event that less than 4 non-toxic dose levels are achieved in the study, the results will be evaluated with the Sponsor and it will be decided if the affected portion of the study will be repeated with an appropriate change in dose levels. If a repeat is required, this will be documented by a study plan amendment.

Means and standard deviations for appropriate dose levels will be calculated and presented in the report together with laboratory historical control information. The mean number of revertant colonies for all treatment groups will be compared with those obtained for the concurrent negative control level.

Colony counts and mean colony counts will generally be reported to the nearest integer value. Where a validated data acquisition and reporting system is in use, the reporting format of that system will be used. Fold changes against control values will generally be tabulated to 1 decimal place. Fold increases against control may occasionally be reported to 2 decimal places to clarify the conclusion with respect to evaluation criteria.

14.1. Assay acceptance criteria:

Acceptable Negative Control: The number of revertants must be within or close the current historical negative control range of the laboratory.

Acceptable Positive Control: The mean value of a Positive Control for a tester strain must be compatible with those generated in the historical positive control data base and be substantially higher (≥ 2 -fold) than the mean number of revertant colonies for its respective Negative Control.

Acceptable highest dose: Depending on the nature of the Test Item, it should reach the toxicity limit (as per Section 11.2), but not exceed 5000 $\mu\text{g}/\text{plate}$, with at least 5 analyzable concentrations, if available.

In the event that the controls fall slightly outside the normal range (historical or Study Plan), the Study Director will be allowed discretion to accept the results of the experiment as valid based on the biological significance.

14.2. Interpretation of Results

Biological relevance of the results will be considered first.

Negative result (no evidence of genotoxicity) is concluded if there is no substantial increase (< 2-fold) in the number of colonies per plate in comparison to the concurrent Negative Control and the data is within or close to the 98% tolerance limit of the Negative Historical Control data. A negative result indicates that the Test Item is non-mutagenic in *S. typhimurium* and *E. coli*.

Positive result (evidence of genotoxicity) is concluded if there is a substantial increase (≥ 2 -fold) in the number of colonies per plate in comparison to the concurrent Negative Control, the mean values are above the 98% tolerance limit of the Negative Historical Control data and a

concentration-related increase over the exposure range tested is obtained. A positive result indicates that the Test Item induces point mutations in *S. typhimurium* and/or *E. coli*.

Equivocal result is concluded if no definite judgment can be made to fit the above criteria. An equivocal result indicates that a definitive conclusion cannot be made by performing the bacterial reverse mutation assay under the conditions described in this study plan. Alternate testing conditions (e.g., a narrower dose interval with the appropriate strain) may be used as an aid in evaluating the test results. Any additional testing or analysis will be approved by the sponsor and will be documented by study plan amendment.

15. COMPUTERIZED SYSTEMS

The following proposed critical computerized systems may be used in the study. The actual critical computerized systems used will be specified in the Final Report.

Data for parameters not required by Study Plan, which are automatically generated by analytical devices used will be retained on file but not reported. Statistical analysis results that are generated by the program but are not required by Study Plan and/or are not scientifically relevant will be retained on file but will not be included in the tabulations.

Proposed Critical Computerized Systems

System Name	Description of Data Collected and/or Analyzed
Sorcerer (Perceptive Instruments)	Revertant colony counts.
Excel (Microsoft)	Descriptive statistics of revertant colony counts data.
Mesa Laboratories AmegaView CMS	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO ₂ , as appropriate.
Johnson Controls Metasys	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories.
Empower 3 (Waters Corporation)	Dose formulation analyses using HPLC.
Empower 3 and Microsoft Excel	Regression analysis and descriptive statistics for dose formulation analytical data.

16. AMENDMENTS AND DEVIATIONS

Changes to the approved Study Plan shall be made in the form of an amendment, which will be signed and dated by the Study Director. Every reasonable effort will be made to discuss any necessary Study Plan changes in advance with the Sponsor.

All Study Plan and SOP deviations will be documented in the study records. The Study Director will notify the Sponsor of deviations that may result in a significant impact on the study as soon as possible.

17. RETENTION OF RECORDS

All study-specific raw data, electronic data, documentation, study plan, retained samples, and interim (if applicable) and final reports will be archived by no later than the date of final report issue. All materials generated by Charles River from this study will be transferred to a Charles

River archive. One year after issue of the unaudited draft report, the Sponsor will be contacted to determine the disposition of materials associated with the study.

18. REPORTING

A comprehensive unaudited Draft Report will be prepared following completion of the study and will be finalized following consultation with the Sponsor. The report will include all information necessary to provide a complete and accurate description of the experimental methods and results and any circumstances that may have affected the quality or integrity of the study.

The Sponsor will receive an electronic version of the unaudited Draft Report provided in Microsoft Word format and an electronic version of the Final Report provided in Adobe Acrobat PDF format (hyperlinked and searchable). The PDF document will be created from native electronic files to the every possible extent, including text and tables generated by the Test Facility. Report components not available in native electronic files and/or original signature pages will be scanned and converted to PDF image files for incorporation.

Reports should be finalized within 6 months of issuance of the unaudited Draft Report. If the Sponsor has not provided comments to the report within 6 months of draft issuance, the report will be finalized by the Test Facility unless other arrangements are made by the Sponsor.

AMENDMENT APPROVAL

(b) (6)

Date: 13 Sep 2016

DEVIATIONS

All deviations that occurred during the study have been authorized/acknowledged by the Study Director, assessed for impact, and documented in the study records. Only minor SOP deviations that did not impact the quality or integrity of the study occurred during the course of the study.

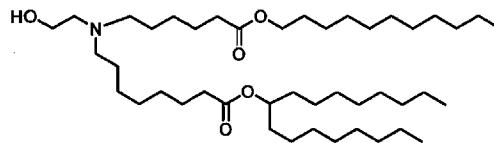
Appendix 2

Test Item Characterization



Certificate of Analysis – Revision 01

Product: MOD-201MS



Chemical Formula: $C_{44}H_{87}NO_5$
Molecular Weight: 710.18

Corden Pharma Sample Name: LP-04-285
Lot #: RL-100-211-1

Tests	Specifications	Results
Appearance	Clear oil, no foreign particulate matter visible	Clear oil, no foreign particulate matter visible
Color	Colorless to yellowish	Yellowish
Solubility (1w% in Ethanol)	Clear solution	Clear solution
Identity		
HPLC-CAD	Retention time of the main peak corresponds to reference	Retention time of the main peak corresponds to reference
IR-Spectrum	Comparable to reference	Comparable to reference
1H -NMR	Conforms to reference	Conforms to reference
MS	M = 709.7 ± 1.0 m/z M + H ⁺ = 710.7 ± 1.0 m/z	Positive ionization mode: M + H ⁺ = 711.07 m/z
Purity		
Purity by CAD-HPLC (% Area)	Report result, target: $\geq 95\%$ Report all imp. $\geq 0.05\%$	MOD-201MS purity = 95.72% Impurities $\geq 0.05\%$: Unk. RRT: 0.66 0.20% Compound 7 RRT: 0.85 0.43% Unk. RRT: 0.93 0.07% Compound 11 RRT: 0.93 <0.05% Unk. RRT: 1.02 1.01% Unk. RRT: 1.03 0.21% Unk. RRT: 1.03 0.26% Unk. RRT: 1.04 0.29% Unk. RRT: 1.05 0.18% Unk. RRT: 1.05 0.26% Unk. RRT: 1.06 0.43% Unk. RRT: 1.06 0.12% Unk. RRT: 1.08 0.06% Unk. RRT: 1.08 0.15% Unk. RRT: 1.09 0.16% Unk. RRT: 1.10 0.07% Unk. RRT: 1.11 0.28%



		Unk. RRT: 1.15 Unk. RRT: 1.18	0.05% 0.07%
Assay			
Assay (solvent free, anhydrous, inorganics free) (%w/w)**	Report result	95.48 %	
Water content, KF (% w/w)	≤ 1.0%	0.2%	
Peroxides	Report result	1.5 mmol/Kg*	
Residual solvents, HS-GC-FID	Acetonitrile ≤ 410 ppm Cyclopentyl-methyl ether ≤ 150 ppm n-Hexane ≤ 290 ppm Dichloromethane ≤ 600 ppm Ethanol ≤ 5000 ppm Heptane ≤ 5000 ppm Methanol ≤ 3000 ppm Ethyl acetate ≤ 5000 ppm	Acetonitrile < 100 ppm Cyclopentyl-methyl ether < 100 ppm n-Hexane < 100 ppm Dichloromethane 400 ppm Ethanol < 100 ppm Heptane < 100 ppm Methanol < 100 ppm Ethyl acetate < 100 ppm	
Elemental impurities, ICP	Cd ≤ 0.2 ppm Pb ≤ 0.5 ppm As ≤ 1.5 ppm Hg ≤ 0.3 ppm Co ≤ 0.5 ppm V ≤ 1 ppm Ni ≤ 2 ppm Li ≤ 25 ppm Sb ≤ 9 ppm Cu ≤ 30 ppm Mg report result Si report result	Cd < 0.1 ppm Pb 0.1 ppm As < 0.1 ppm Hg < 0.1 ppm Co < 0.1 ppm V < 0.1 ppm Ni 0.1 ppm Li < 0.1 ppm Sb < 0.1 ppm Cu 0.1 ppm Mg 0.5 ppm Si 21.8 ppm	
Residue on ignition (%w/w) USP <281>	≤ 0.1 %	< 0.1%	

* Note: Hydroperoxide content 1 mmol/Kg = 34 ppm

**Assay = 100% - HPLC impurities (%) – water (%) – inorganics (%)

Recommended Storage: -20 °C ± 5 °C, Protect from light, moisture and air. Store under inert gas.

Date: October 27, 2016

Signature:

(b) (6)

Appendix 3
Historical Negative and Positive Control Results

Historical Negative Control Results

Strain	S9	Mean	SD	Range	98% Tolerance Limit
TA1535	0	16	4	8 - 29	26
TA1537	0	13	3	6 - 18	18
TA98	0	32	6	20 - 58	47
TA100	0	119	19	78 - 185	160
WP2 <i>uvrA</i>	0	42	7	28 - 67	59
TA1535	+	15	4	7 - 28	27
TA1537	+	15	3	8 - 22	21
TA98	+	42	7	20 - 63	57
TA100	+	137	19	98 - 185	178
WP2 <i>uvrA</i>	+	50	8	34 - 76	69

SD Standard deviation

Historical Positive Control Results

Strain	Treatment	Dose (µg/plate)	S9	Mean	SD	Range
TA1535	NaAz	0.5	0	395	41	268 - 478
TA1537	9AC	50	0	538	250	219 - 1334
TA98	2NF	1	0	152	19	106 - 198
TA100	NaAz	0.5	0	511	45	413 - 628
WP2 <i>uvrA</i>	NQO	0.5	0	216	69	85 - 489
TA1535	2AA	5	+	241	42	139 - 356
TA1537	BaP	5	+	148	33	68 - 225
TA98	BaP	5	+	527	80	332 - 695
TA100	BaP	5	+	1066	128	754 - 1465
WP2 <i>uvrA</i>	2AA	20	+	207	46	137 - 361

SD Standard deviation

Means of the mean results from non-GLP and QA-audited GLP studies that occurred from January 2014 to May 2016, using the plate incorporation method.

Appendix 4
Dose Formulation Analysis Report



FINAL REPORT

Study Phase: Analytical Chemistry

Test Facility Study No. 9601567

TEST FACILITY:
Charles River Laboratories Montreal ULC

(b) (6)

A large rectangular area of the document is redacted, indicated by a solid gray box. The text "(b) (6)" is positioned at the top left of this redacted area.

TABLE OF CONTENTS

LIST OF TABLES3

LIST OF APPENDICES3

1. SUMMARY4

2. INTRODUCTION4

3. EXPERIMENTAL DESIGN4

4. MATERIALS AND METHODS4

4.1. Materials4

4.1.1. Reference Standard4

4.1.2. Characterization of Reference Standard4

4.1.3. Inventory and Disposition of Reference Standard4

4.2. Methods.....5

4.2.1. Analytical Procedure.....5

4.3. Computerized Systems.....5

5. RESULTS AND DISCUSSIONS5

6. CONCLUSION5

7. REPORT APPROVAL.....6

LIST OF TABLES

Table 1 Study Samples – Concentration7

LIST OF APPENDICES

Appendix 1 Analytical Procedure.....8
Appendix 2 Certificate of Analysis15

1. SUMMARY

Dose formulation samples have been analyzed by High Performance Liquid Chromatography (HPLC) for the determination of SM-102.

The dose formulations were within specification.

2. INTRODUCTION

This report describes the analytical evaluation of SM-102 in dose formulations (Ethanol) from Study No. 9601567.

For the work detailed in this report, the analytical phase experimental start date was 12 Sep 2016, and the analytical phase experimental completion date was 13 Sep 2016.

3. EXPERIMENTAL DESIGN

Analysis of dose formulations was carried out with regard to concentration.

During the study, duplicate samples were taken from each Test Item formulation (Dose Number 1 to 8). The samples were analyzed on the day of preparation.

4. MATERIALS AND METHODS

4.1. Materials

4.1.1. Reference Standard

Identification:	SM-102
Lot No.:	RL-100-211-1
Purity (Assay):	95.48% (95.3% used for calculations)
Retest Date:	27 Oct 2017
Storage Conditions:	Kept in a freezer set to maintain -20°C
Supplier:	Moderna Therapeutics, Inc.

4.1.2. Characterization of Reference Standard

The Supplier provided the documentation for the identity, strength, purity, composition, and stability for the reference standard. A copy of the supplied Certificate of Analysis or equivalent documentation are presented in [Appendix 2](#).

4.1.3. Inventory and Disposition of Reference Standard

Records of the receipt, distribution, and storage of the reference standard was maintained. All unused Sponsor-supplied reference standard was returned to the Sponsor.

4.2. Methods

4.2.1. Analytical Procedure

The method of analysis is documented in analytical procedure AP.9601567.SL.01 ([Appendix 1](#)) and was previously validated under Study No. 1801841. Prior to study commencement, stability data were generated by the department of Analytical Chemistry, Charles River, CR MTL for up to 5 days, for formulations stored at ambient temperature and in a refrigerator set to maintain a temperature of 4°C, respectively, over the concentration range of 0.00900 - 60.0 mg/mL, under Study No. 1801841.

4.3. Computerized Systems

Critical computerized systems used in this study phase are listed below (see [Text Table 1](#)).

Text Table 1
Computerized Systems

System Name	Version No.	Description of Data Collected and/or Analyzed
Empower 3 (Waters Corporation)	Build 3471 SR1	Data acquisition for dose formulation analysis, including regression analysis and measurement of concentration and recovery of dose formulations using HPLC
Mesa Laboratories AmegaView CMS	v3.0 Build 1208.8	Continuous Monitoring System. Monitoring of standalone fridges, freezers, incubators, and selected laboratories to measure temperature, relative humidity, and CO ₂ , as appropriate
Johnson Controls Metasys	MVE 5.4 (M5)	Building Automation System. Control of HVAC and other building systems, as well as temperature/humidity control and trending in selected laboratories and animal rooms

5. RESULTS AND DISCUSSIONS

The results of all samples were found to be within the acceptance criteria of $\pm 10\%$ of their theoretical concentrations for stock solutions (dose number 8) and $\pm 15\%$ of their theoretical concentrations for lower level solutions (dose number 1 to 7), except for dose number 1 to 3 (mean recovery: 120%, 116% and 116%, respectively). The retention samples were not analyzed and as per the Study Directors instruction the dose volumes were adjusted accordingly. Results are presented in [Table 1](#).

All results presented in the tables of the report are calculated using non-rounded values as per the raw data rounding procedure and may not be exactly reproduced from the individual data presented.

6. CONCLUSION

The dose formulations were within specification, except for dose number 1 to 3.

7. REPORT APPROVAL

(b) (6)

Date: 12 Jan 2017

Table 1 Study Samples – Concentration

Preparation Date	Dose Number	Theoretical Concentration		Measured Concentration (ug/mL)	Percent of Theoretical	Mean Measured Concentration (ug/mL)	Mean Percent of Theoretical
		ug/mL	mg/mL				
12 Sep 2016	1	(b) (4)					
	2						
	3						
	4						
	5						
	6						
	7						
	8						

a: out of acceptance criteria

Appendix 1
Analytical Procedure

Analytical Procedure (AP.9601567.SL.01)

Page 1 of 6

High Performance Liquid Chromatographic Method Using Ultraviolet/Visible Detection for the Determination of SM-102 in Dose Formulations

This analytical procedure can be used for the quantitation of SM-102 in dose formulation samples in ethanol over the concentration range of 0.00900 to 60.0 mg/mL.

Reference Standard and Vehicle

Reference Standard	SM-102
Lot number	RL-100-211-1
Purity	95.3% (to be used for calculations)

Vehicle: Ethanol

For storage conditions for Reference Standard, refer to the corresponding log sheets or transfer sheets.

NOTES

- Modifications may be made to the conditions below in order to optimize the chromatography.
- All volumes and weights listed throughout this procedure may be scaled up or down as long as the final concentration remains the same.
- Any changes are to be noted in the raw data.
- Unless otherwise indicated, information relating to the time of mixing/stirring, temperature used in the preparation of solutions, diluents, mobile phases and vehicles will be considered non-critical.
- If a step is deemed critical, it will be noted within the procedure, and a positive entry will be made in the raw data. In addition, the use of sonication to aid in the dissolution of reagents or TL, will also be considered an appropriate method within the laboratory unless it is clearly indicated that sonication is not to be used.
- The analytical method was previously validated under study 1801841.

Chromatographic Conditions

(b) (4)



Analytical Procedure (AP.9601567.SL.01)

Page 2 of 6

Reagents

Unless specified, reagents with grade (A.C.S., USP et al) or numerical purity will be used.

(b) (4)

A rectangular gray box redacting the content of the Reagents section.

Preparation of Solutions

(b) (4)

A large rectangular gray box redacting the content of the Preparation of Solutions section.

Analytical Procedure (AP.9601567.SL.01)

Page 3 of 6

Preparation of Standards

(b) (4)

Table 1: Preparation of standards

STD identification	Stock solution	Volume of stock solution (µL)	Final volume (mL)	Final concentration (µg/mL)
STD A	diluent	(b) (4)		
STD B	STD STK A			
STD C	STD STK B			
STD D	STD STK A			
STD E	STD STK B			
STD F	STD STK A			
STD G	STD STK B			

Preparation of Blanks

(b) (4)

Analytical Procedure (AP.9601567.SL.01)

Page 4 of 6

Preparation of Spikes

(b) (4)



Table 2: Dilution scheme for spiked samples

SPK ID	Replicate	Aliquot of SPK (µL)	Volume of acetonitrile to add (µL)	Flask volume (mL)	Final concentration (µg/mL)	Dilution factor
SPK A	n = 2	(b) (4)				
SPK B	n = 2					

Analytical Procedure (AP.9601567.SL.01)

Page 5 of 6

(b) (4)

Table 3: Dilution scheme for samples

Dose number	Concentration		Initial Dilution			Final concentration (µg/mL)	Dilution factor
	(µg/mL)	(mg/mL)	Aliquot of sample (µL)	Volume of ACN to add (µL)	Flasks volume (mL)		
1	(b) (4)						
2							
3							
4							
5							
6							
7							
8							

Injection Sequence

(b) (4)

Analytical Procedure (AP.9601567.SL.01)

Page 6 of 6

Calculations

System suitability

- Calculate relative standard deviation (%RSD) in response of CAL (n = 3) using the following equation:
$$\% \text{ RSD} = (\text{SD} \div \text{A}) \times 100$$

SD – standard deviation in response
A – average response
- Calculate system stability using the following equation:
$$(\text{A}_2 - \text{A}_1) \div \text{A}_1 \times 100$$

A₁ – average response (n = 3) of CAL at the beginning of the run.
A₂ – average response (n = 3) of CAL at the end of the run.

Standard curve

- Perform the least squares fit regression of peak area versus concentration (type of curve fit: **Linear**; weighting factor: 1/x²)

Calculation of concentrations

- Using Empower 3 Custom Field, calculate concentrations and accuracies of spikes and study samples.

Acceptance Criteria

Unless specified in the study plan or below, refer to SOP CAD-002 and SOP CAD-003 for acceptance criteria.

Study Samples

(b) (4)

AP Version Control

Initial version.

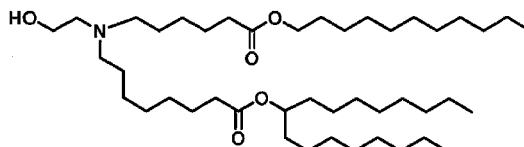
Verified by	(b) (6)	Date	09 Sep 2016
Approved by	(b) (6)	Date	09 Sep 2016
Individual Scientist	(b) (6)		
Authorized by	(b) (6)	Date	09 Sep 2016
Scientific Director	☒		

Appendix 2
Certificate of Analysis



Certificate of Analysis – Revision 01

Product: MOD-201MS



Chemical Formula: $C_{44}H_{87}NO_5$
Molecular Weight: 710.18

Corden Pharma Sample Name: LP-04-285
Lot #: RL-100-211-1

Tests	Specifications	Results
Appearance	Clear oil, no foreign particulate matter visible	Clear oil, no foreign particulate matter visible
Color	Colorless to yellowish	Yellowish
Solubility (1w% in Ethanol)	Clear solution	Clear solution
Identity		
HPLC-CAD	Retention time of the main peak corresponds to reference	Retention time of the main peak corresponds to reference
IR-Spectrum	Comparable to reference	Comparable to reference
1H-NMR	Conforms to reference	Conforms to reference
MS	M = 709.7 ± 1.0 m/z M + H ⁺ = 710.7 ± 1.0 m/z	Positive ionization mode: M + H ⁺ = 711.07 m/z
Purity		
Purity by CAD-HPLC (% Area)	Report result, target: $\geq 95\%$ Report all imp. $\geq 0.05\%$	MOD-201MS purity = 95.72% Impurities $\geq 0.05\%$: Unk. RRT: 0.66 0.20% Compound 7 RRT: 0.85 0.43% Unk. RRT: 0.93 0.07% Compound 11 RRT: 0.93 <0.05% Unk. RRT: 1.02 1.01% Unk. RRT: 1.03 0.21% Unk. RRT: 1.03 0.26% Unk. RRT: 1.04 0.29% Unk. RRT: 1.05 0.18% Unk. RRT: 1.05 0.26% Unk. RRT: 1.06 0.43% Unk. RRT: 1.06 0.12% Unk. RRT: 1.08 0.06% Unk. RRT: 1.08 0.15% Unk. RRT: 1.09 0.16% Unk. RRT: 1.10 0.07% Unk. RRT: 1.11 0.28%



		Unk. RRT: 1.15 0.05% Unk. RRT: 1.18 0.07%
Assay		
Assay (solvent free, anhydrous, inorganics free) (%w/w)**	Report result	95.48 %
Water content, KF (% w/w)	≤ 1.0%	0.2%
Peroxides	Report result	1.5 mmol/Kg*
Residual solvents, HS-GC-FID	Acetonitrile ≤ 410 ppm Cyclopentyl-methyl ether ≤ 150 ppm n-Hexane ≤ 290 ppm Dichloromethane ≤ 600 ppm Ethanol ≤ 5000 ppm Heptane ≤ 5000 ppm Methanol ≤ 3000 ppm Ethyl acetate ≤ 5000 ppm	Acetonitrile < 100 ppm Cyclopentyl-methyl ether < 100 ppm n-Hexane < 100 ppm Dichloromethane 400 ppm Ethanol < 100 ppm Heptane < 100 ppm Methanol < 100 ppm Ethyl acetate < 100 ppm
Elemental impurities, ICP	Cd ≤ 0.2 ppm Pb ≤ 0.5 ppm As ≤ 1.5 ppm Hg ≤ 0.3 ppm Co ≤ 0.5 ppm V ≤ 1 ppm Ni ≤ 2 ppm Li ≤ 25 ppm Sb ≤ 9 ppm Cu ≤ 30 ppm Mg report result Si report result	Cd < 0.1 ppm Pb 0.1 ppm As < 0.1 ppm Hg < 0.1 ppm Co < 0.1 ppm V < 0.1 ppm Ni 0.1 ppm Li < 0.1 ppm Sb < 0.1 ppm Cu 0.1 ppm Mg 0.5 ppm Si 21.8 ppm
Residue on ignition (%w/w) USP <281>	≤ 0.1 %	< 0.1%

* Note: Hydroperoxide content 1 mmol/Kg = 34 ppm

**Assay = 100% - HPLC impurities (%) – water (%) – inorganics (%)

Recommended Storage: -20 °C ± 5 °C, Protect from light, moisture and air. Store under inert gas.

Date: October 27, 2016

Signature:

(b) (6) (h) (6)